

```
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

The screenshot displays a SQL IDE interface. The top pane shows a SQL query being executed: `CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name VARCHAR(50), transaction_type VARCHAR(20), amount DECIMAL(10,2), transaction_date DATE );`. The bottom pane shows the 'Action Output' table with the following data:

#	Time	Action	Response	Duration / Fetch Time
98	14:25:52	CREATE TABL...	0 row(s) affected	0.041 sec

Below the output table, a list of queries is shown, including:

- DROP DATABASE bookflow_db;
- CREATE DATABASE IF NOT EXISTS bookflow_db;
- USE bookflow_db;
- CREATE TABLE bank_transactions (
 txn_id INT PRIMARY KEY,
 customer_name VARCHAR(50),
 branch_name VARCHAR(50),
 transaction_type VARCHAR(20),
 amount DECIMAL(10,2),
 transaction_date DATE
);

The 'Output' section at the bottom shows the execution results for these queries, with the fourth query (CREATE TABLE) being the most recent and highlighted.

Step 2 — ALTER TABLE — ADD COLUMN

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

ator:.....

Query 1 x

Limit to 1000 rows

```

4 • CREATE TABLE bank_transactions (
5     txn_id INT PRIMARY KEY,
6     customer_name VARCHAR(50),
7     branch_name VARCHAR(50),
8     transaction_type VARCHAR(20),
9     amount DECIMAL(10,2),
10    transaction_date DATE
11 );
12 • ALTER TABLE bank_transactions
13     ADD account_no VARCHAR(20);

```

Output

Action Output

#	Time	Action	
✓ 1	19:38:21	DROP DATABASE bookflow_db	4
✓ 2	19:38:21	CREATE DATABASE IF NOT EXISTS bookflow_db	1
✓ 3	19:38:21	USE bookflow_db	0
✓ 4	19:38:31	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name ...	0
✓ 5	19:40:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0

istration Schemas

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Step 3 — ALTER TABLE — MODIFY COLUMN

```
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

ator:.....

Query 1 x

Limit to 1000 rows

```

6     customer_name VARCHAR(50),
7     branch_name VARCHAR(50),
8     transaction_type VARCHAR(20),
9     amount DECIMAL(10,2),
10    transaction_date DATE
11 );
12 • ALTER TABLE bank_transactions
13     ADD account_no VARCHAR(20);
14 • ALTER TABLE bank_transactions
15     MODIFY customer_name VARCHAR(100);

```

Output

Action Output

#	Time	Action	Me
✓ 1	19:38:21	DROP DATABASE bookflow_db	4 rc
✓ 2	19:38:21	CREATE DATABASE IF NOT EXISTS bookflow_db	1 rc
✓ 3	19:38:21	USE bookflow_db	0 rc
✓ 4	19:38:31	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name ...	0 rc
✓ 5	19:40:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 rc
✓ 6	19:47:25	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 rc

istration Schemas

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Step 4 — TRUNCATE TABLE

```
TRUNCATE TABLE bank_transactions;
```

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Database' selected. The right pane shows the 'Query 1' editor with the following SQL script:

```
7  branch_name VARCHAR(50),
8  transaction_type VARCHAR(20),
9  amount DECIMAL(10,2),
10 transaction_date DATE
11 );
12 • ALTER TABLE bank_transactions
13   ADD account_no VARCHAR(20);
14 • ALTER TABLE bank_transactions
15   MODIFY customer_name VARCHAR(100);
16 • TRUNCATE TABLE bank_transactions;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action
1	19:38:21	DROP DATABASE bookflow_db
2	19:38:21	CREATE DATABASE IF NOT EXISTS bookflow_db
3	19:38:21	USE bookflow_db
4	19:38:31	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name ...
5	19:40:32	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)
6	19:47:25	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)
7	19:49:59	TRUNCATE TABLE bank_transactions

Step 5 — DROP TABLE

```
DROP TABLE bank_backup;
```

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with 'Database' selected. The right pane shows the 'Query 1' editor with the following SQL script:

```
8  transaction_type VARCHAR(20),
9  amount DECIMAL(10,2),
10 transaction_date DATE
11 );
12 • ALTER TABLE bank_transactions
13   ADD account_no VARCHAR(20);
14 • ALTER TABLE bank_transactions
15   MODIFY customer_name VARCHAR(100);
16 • TRUNCATE TABLE bank_transactions;
17 • DROP TABLE bank_transactions;
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action
1	19:52:25	DROP DATABASE bookflow_db
2	19:52:25	CREATE DATABASE IF NOT EXISTS bookflow_db
3	19:52:25	USE bookflow_db
4	19:52:25	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch ...
5	19:52:25	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)
6	19:52:25	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)
7	19:52:25	TRUNCATE TABLE bank_transactions
8	19:52:25	DROP TABLE bank_transactions

Step 6 — Insert Sample Data (10 rows)

[illegible]

Step 7 — INSERT

```
INSERT INTO bank_transactions
VALUES
(111, 'Preetish', 'Vizag', 'Deposit', 7000, '2024-01-18', 'ACC111');
```

The screenshot shows the SQL Developer interface. The 'Query 1' window contains the following SQL commands:

```
31 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
32
33 • SELECT * FROM bank_transactions;
34 • INSERT INTO bank_transactions
35 VALUES (111, 'Preetish', 'Vizag', 'Deposit', 7000, '2024-01-18', 'ACC111');
36
```

The 'Result Grid' shows the following data:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	NULL
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

The 'Output' window shows the following actions:

#	Time	Action	Message
31	20:57:45	TRUNCATE TABLE bank_transactions	0 row(s)
32	20:57:45	CREATE TABLE bank_backup (id int)	0 row(s)
33	20:57:45	DROP TABLE bank_backup	0 row(s)
34	20:57:45	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16', 'ACC111');	10 row(s)
35	20:57:45	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s)
36	20:57:45	INSERT INTO bank_transactions VALUES (111, 'Preetish', 'Vizag', 'Deposit', 7000, '2024-01-18', 'ACC111');	1 row(s)

Step 8 — UPDATE

```
UPDATE bank_transactions
SET amount = 5000
WHERE txn_id = 105;
```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

Query 1 x

```

34 • INSERT INTO bank_transactions
35 VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111');
36 • UPDATE bank_transactions SET amount = 5000
37 WHERE txn_id = 105;
38

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	HULL
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	HULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	HULL
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	HULL
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	HULL
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	HULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	HULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	HULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	HULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	HULL

bank_transactions 7 x

Output

Action Output

#	Time	Action	Message
6	20:59:55	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected
7	20:59:55	TRUNCATE TABLE bank_transactions	0 row(s) affected
8	20:59:55	CREATE TABLE bank_backup (id int)	0 row(s) affected
9	20:59:55	DROP TABLE bank_backup	0 row(s) affected
10	20:59:55	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (101, 'Abhi', 'Hyderabad', 'Deposit', 5000, '2024-01-05', 'ACC001');	10 row(s) affected
11	20:59:55	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) affected
12	20:59:55	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111');	1 row(s) affected
13	20:59:55	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected

Step 9 — DELETE

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

Query 1 x

```

18 (102,'Arjun','Hyderabad','Withdrawal',2000,'2024-01-06','ACC002');
19 (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08','ACC003');
20 (104,'Sanjay','Vizag','Deposit',8000,'2024-01-10','ACC004');
21 (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11','ACC005');
22 (106,'Harshitha','Vizag','Deposit',15000,'2024-01-12','ACC006');
23 (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13','ACC007');
24 (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14','ACC008');
25 (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15','ACC009');
26 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16','ACC010');
27 • INSERT INTO bank_transactions VALUES
28 (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111');
29 • UPDATE bank_transactions SET amount = 5000
30 WHERE txn_id = 105;
31 • DELETE FROM bank_transactions
32 WHERE txn_id = 111;
33
34
35

```

Output

Action Output

#	Time	Action	Message
16	09:36:08	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected
17	09:36:08	TRUNCATE TABLE bank_transactions	0 row(s) affected
18	09:36:08	DROP TABLE IF EXISTS bank_backup	0 row(s) affected
19	09:36:08	INSERT INTO bank_transactions VALUES (101,'Abhi','Hyderabad','Deposit',5000,'2024-01-05','ACC001'), (102,'Arjun','Hyderabad','Withdrawal',2000,'2024-01-06','ACC002');	10 row(s) affected
20	09:36:08	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111');	1 row(s) affected
21	09:36:08	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected
22	09:36:08	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected

Step 10 — SELECT (All Rows)

```
SELECT *  
FROM bank_transactions;
```

The screenshot shows a database management tool interface with a menu bar (Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar contains a tree view with categories like 'Management', 'Performance', and 'Administration'. The main area displays a SQL query in a text editor:

```
35 VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111');  
36 UPDATE bank_transactions SET amount = 5000  
37 WHERE txn_id = 105;  
38 SELECT *  
39 FROM bank_transactions;
```

Below the query editor, the 'Result Grid' shows the output of the query. It is a table with 8 columns: txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, and account_no. The data is as follows:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
111	Preetish	Vizag	Deposit	7000.00	2024-01-18	ACC111
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Below the result grid, the 'Output' section shows a list of actions performed by the tool, including truncating the table, creating a backup, dropping the backup, inserting data, updating the amount, and selecting all rows.

#	Time	Action
7	20:59:55	TRUNCATE TABLE bank_transactions
8	20:59:55	CREATE TABLE bank_backup (id int)
9	20:59:55	DROP TABLE bank_backup
10	20:59:55	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (103, 'Kiran', 'Vijayawada', 'Deposit', 12000.00, '2024-01-08', NULL)
11	20:59:55	SELECT * FROM bank_transactions LIMIT 0, 1000
12	20:59:55	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111')
13	20:59:55	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
14	21:01:44	SELECT * FROM bank_transactions LIMIT 0, 1000

Step 11 — SELECT WITH WHERE

```
SELECT *
FROM bank_transactions
WHERE transaction_type='Deposit';
```

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
39 FROM bank_transactions;
40 SELECT *
41 FROM bank_transactions
42 WHERE transaction_type='Deposit';
43
44
45
```

The Results tab displays the output of the query in a table format:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	NA
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NA
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	NA
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	NA
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NA
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NA
111	Preetish	Vizag	Deposit	7000.00	2024-01-18	ACC111

The Output tab shows the execution log with the following actions:

```
9 20:59:55 DROP TABLE bank_backup
10 20:59:55 INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (101, 'Abhi', 'Hyderabad', 'Deposit', 5000.00, '2024-01-05', 'NA')
11 20:59:55 SELECT * FROM bank_transactions LIMIT 0, 1000
12 20:59:55 INSERT INTO bank_transactions VALUES (111, 'Preetish', 'Vizag', 'Deposit', 7000.00, '2024-01-18', 'ACC111')
13 20:59:55 UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
14 21:01:44 SELECT * FROM bank_transactions WHERE transaction_type='Deposit'
15 21:03:10 SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
```

Step 12 — SELECT WITH ORDER BY

```
SELECT *
FROM bank_transactions
ORDER BY amount DESC;
```

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
39 FROM bank_transactions;
40 SELECT *
41 FROM bank_transactions
42 WHERE transaction_type='Deposit';
43 SELECT *
44 FROM bank_transactions ORDER BY amount DESC;
45
```

The Results tab displays the output of the query in a table format:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NA
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	NA
111	Preetish	Vizag	Deposit	7000.00	2024-01-18	ACC111
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	NA
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NA
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NA
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	NA
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NA

The Output tab shows the execution log with the following actions:

```
10 20:59:55 INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (101, 'Abhi', 'Hyderabad', 'Deposit', 5000.00, '2024-01-05', 'NA')
11 20:59:55 SELECT * FROM bank_transactions LIMIT 0, 1000
12 20:59:55 INSERT INTO bank_transactions VALUES (111, 'Preetish', 'Vizag', 'Deposit', 7000.00, '2024-01-18', 'ACC111')
13 20:59:55 UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
14 21:01:44 SELECT * FROM bank_transactions WHERE transaction_type='Deposit'
15 21:03:10 SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
16 21:05:59 SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
```

Step 12 — SELECT WITH ORDER BY

```
SELECT *  
FROM bank_transactions  
ORDER BY amount DESC;
```

The screenshot shows the SQL Enterprise Manager interface. The query editor displays the following SQL statement:

```
41 SELECT *  
42 FROM bank_transactions  
43 ORDER BY amount DESC;  
44  
45  
46
```

The Results pane shows a table with 7 columns: txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, and account_no. The data is sorted by amount in descending order.

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

The Action Output pane shows the following actions:

#	Time	Action	Message
9	09:40:15	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111')	1 row
10	09:40:15	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row
11	09:40:15	DELETE FROM bank_transactions WHERE txn_id = 111	1 row
12	09:40:15	SELECT * FROM bank_transactions LIMIT 0, 1000	10 rows
13	09:40:15	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	6 rows
14	09:40:15	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 rows
15	09:40:15	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 rows

SECTION 3 — DCL (Data Control Language)

Step 13 — GRANT SELECT

```
GRANT SELECT  
ON bank_transactions  
TO Auditor1;
```

The screenshot shows the SQL Enterprise Manager interface. The query editor displays the following SQL statement:

```
39 FROM bank_transactions  
40 ORDER BY amount DESC;  
41 SELECT *  
42 FROM bank_transactions  
43 ORDER BY amount DESC;  
44 GRANT SELECT ON bank_transactions.* TO 'Auditor1'@'localhost';
```

The Results pane shows a table with 7 columns: txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, and account_no. The data is sorted by amount in descending order.

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

The Action Output pane shows the following actions:

#	Time	Action	Message
8	10:06:38	INSERT INTO bank_transactions VALUES (101,'Abhi','Hyderabad','Deposit',5000,'2024-01-05','ACC001')	1 row
9	10:06:38	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111')	1 row
10	10:06:38	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row
11	10:06:38	DELETE FROM bank_transactions WHERE txn_id = 111	1 row
12	10:06:38	SELECT * FROM bank_transactions LIMIT 0, 1000	10 rows
13	10:06:38	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	6 rows
14	10:06:38	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 rows
15	10:06:38	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 rows

Step 14 — GRANT ALL PRIVILEGES

```
GRANT ALL  
ON bank_transactions  
TO BranchManager;
```

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Edit View Query Database Server Tools Scripting Help

itor:.....

Query 1 x

Limit to 1000 rows

```
39 FROM bank_transactions  
40 ORDER BY amount DESC;  
41 • SELECT *  
42 FROM bank_transactions  
43 ORDER BY amount DESC;  
44 • CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

bank_transactions 21 bank_transactions 22 bank_transactions 23 bank_transactions 24 x

Output

Action Output

#	Time	Action
13	10:09:02	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
14	10:09:02	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
15	10:09:02	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
16	10:09:02	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'
17	10:09:02	GRANT SELECT ON bank_transactions.* TO 'Auditor1'@'localhost'
18	10:09:02	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'
19	10:09:02	GRANT ALL PRIVILEGES ON bank_transactions.* TO 'BranchManager'@'localhost'

Step 15 — REVOKE SELECT

```
REVOKE SELECT  
ON bank_transactions  
FROM Auditor1;
```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

Query 1 x

```

44 • CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234';
45 • GRANT SELECT ON bank_transactions.* TO 'Auditor1'@'localhost';
46 • CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234';
47 • GRANT ALL PRIVILEGES ON bank_transactions.* TO 'BranchManager'@'localhost';
48 • REVOKE SELECT ON bank_transactions.*
49   FROM 'Auditor1'@'localhost';

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

bank_transactions 37 bank_transactions 38 bank_transactions 39 bank_transactions 40 x

Output

Action Output

#	Time	Action
✓ 56	10:12:40	INSERT INTO bank_transactions VALUES (101,'Abhi','Hyderabad','Deposit',5000,'2024-01-05','ACC001'). (10.
✓ 57	10:12:40	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111)
✓ 58	10:12:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
✓ 59	10:12:40	DELETE FROM bank_transactions WHERE txn_id = 111
✓ 60	10:12:40	SELECT * FROM bank_transactions LIMIT 0, 1000
✓ 61	10:12:40	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
✓ 62	10:12:40	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
✓ 63	10:12:40	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000

Info Session

Step 16 — REVOKE ALL

```

REVOKE ALL
ON bank_transactions
FROM BranchManager;

```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

Query 1 x

```

44 • CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234';
45 • GRANT SELECT ON bank_transactions.* TO 'Auditor1'@'localhost';
46 • CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234';
47 • GRANT ALL PRIVILEGES ON bank_transactions.* TO 'BranchManager'@'localhost';
48 • REVOKE SELECT ON bank_transactions.*
49   FROM 'Auditor1'@'localhost';

```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

bank_transactions 45 bank_transactions 46 bank_transactions 47 bank_transactions 48 x

Output

Action Output

#	Time	Action
✓ 8	10:15:41	INSERT INTO bank_transactions VALUES (101,'Abhi','Hyderabad','Deposit',5000,'2024-01-05','ACC001'). (10.
✓ 9	10:15:41	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111)
✓ 10	10:15:41	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
✓ 11	10:15:41	DELETE FROM bank_transactions WHERE txn_id = 111
✓ 12	10:15:41	SELECT * FROM bank_transactions LIMIT 0, 1000
✓ 13	10:15:41	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
✓ 14	10:15:41	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
✓ 15	10:15:41	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000

Info Session

SECTION 4 — TCL (Transaction Control Language)

Step 17 — COMMIT

```
UPDATE bank_transactions
SET amount=6000
WHERE txn_id=101;

COMMIT;
```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

SQL

ator:.....

Query 1 x

Limit to 1000 rows

40 ORDER BY amount DESC;

41 • SELECT *

42 FROM bank_transactions

43 ORDER BY amount DESC;

44 • CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234';

45 • GRANT SELECT ON bank_transactions.* TO 'Auditor1'@'localhost';

46 • CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234';

47 • GRANT ALL PRIVILEGES ON bank_transactions.* TO 'BranchManager'@'localhost';

48 • REVOKE SELECT ON bank_transactions.*

49 FROM 'Auditor1'@'localhost';

50 • REVOKE ALL PRIVILEGES

51 ON bank_transactions.*

52 FROM 'BranchManager'@'localhost';

53 • UPDATE bank_transactions SET amount=6000

54 WHERE txn_id=101;

55

56 • COMMIT;

57

Management

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

Management

- Startup / Shutdown
- Server Logs
- Options File

Management

- Dashboard
- Performance Reports
- Performance Schema Setup

Illustration Schemas

ation:.....

object selected

Output

Action Output

#	Time	Action
✓ 1	10:17:08	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101
✓ 2	10:17:08	COMMIT

Step 18 — SAVEPOINT

```
SAVEPOINT Before_Update;
```

The screenshot shows the MySQL Workbench interface. The left sidebar contains a tree view with categories like 'MANAGEMENT', 'INCE', and 'PERFORMANCE'. The main window displays a SQL query in the 'Query 1' tab. The query is as follows:

```
40 ORDER BY amount DESC;  
41 • SELECT *  
42 FROM bank_transactions  
43 ORDER BY amount DESC;  
44 • CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'  
45 • GRANT SELECT ON bank_transactions.* TO 'Auditor1'@'loca  
46 • CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '  
47 • GRANT ALL PRIVILEGES ON bank_transactions.* TO 'BranchM  
48 • REVOKE SELECT ON bank_transactions.*  
49 FROM 'Auditor1'@'localhost';  
50 • REVOKE ALL PRIVILEGES  
51 ON bank_transactions.*  
52 FROM 'BranchManager'@'localhost';  
53 • UPDATE bank_transactions SET amount=6000  
54 WHERE txn_id=101;  
55  
56 • COMMIT;  
57 • SAVEPOINT Before_Update;
```

Below the query editor, the 'Output' tab is selected, showing the 'Action Output' table:

#	Time	Action
1	10:17:08	UPDATE bank_transactions SET amount=6000 WHERE txn_id=101
2	10:17:08	COMMIT
3	11:57:10	SAVEPOINT Before_Update

Step 19 — ROLLBACK TO SAVEPOINT

```
UPDATE bank_transactions  
SET amount=99999  
WHERE txn_id=102;  
  
ROLLBACK TO Before_Update;  
  
SELECT * FROM bank_transactions WHERE txn_id=102;
```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

ator:.....

Query 1 x

Limit to 1000 rows

57 • SAVEPOINT Before_Update;
58 • UPDATE bank_transactions SET amount=99999
59 • WHERE txn_id=102; ROLLBACK TO Before_Update;
60 • SELECT * FROM bank_transactions WHERE txn_id=102;
61 • COMMIT;
62

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

bank_transactions 49 bank_transactions 50 bank_transactions 51 bank_transactions 52 x

Output

Action Output

#	Time	Action
8	11:59:28	INSERT INTO bank_transactions VALUES (101,'Abhi','Hyderabad','Deposit',5000,'2024-01-05','ACC001)
9	11:59:28	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111)
10	11:59:29	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
11	11:59:29	DELETE FROM bank_transactions WHERE txn_id = 111
12	11:59:29	SELECT * FROM bank_transactions LIMIT 0, 1000
13	11:59:29	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
14	11:59:29	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
15	11:59:29	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000

Info Session

Step 20 — Full TCL Example

```
UPDATE bank_transactions
SET amount=7000
WHERE txn_id=101;

SAVEPOINT SP1;

UPDATE bank_transactions
SET amount=9000
WHERE txn_id=102;

ROLLBACK TO SP1;

COMMIT;
```

2520090088_B.DHARSHAN x

Edit View Query Database Server Tools Scripting Help

ator:.....

Query 1 x

Limit to 1000 rows

56 • COMMIT;
57 • SAVEPOINT Before_Update;
58 • UPDATE bank_transactions SET amount=99999
59 • WHERE txn_id=102; ROLLBACK TO Before_Update;
60 • SELECT * FROM bank_transactions WHERE txn_id=102;
61 • COMMIT;
62

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Harshitha	Vizag	Deposit	15000.00	2024-01-12	ACC006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC008
104	Sanjay	Vizag	Deposit	8000.00	2024-01-10	ACC004
101	Abhi	Hyderabad	Deposit	5000.00	2024-01-05	ACC001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC009
102	Arjun	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC007

bank_transactions 53 bank_transactions 54 bank_transactions 55 bank_transactions 56 x

Output

Action Output

#	Time	Action
28	12:00:40	INSERT INTO bank_transactions VALUES (101,'Abhi','Hyderabad','Deposit',5000,'2024-01-05','ACC001)
29	12:00:40	INSERT INTO bank_transactions VALUES (111,'Preetish','Vizag','Deposit',7000,'2024-01-18','ACC111)
30	12:00:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105
31	12:00:40	DELETE FROM bank_transactions WHERE txn_id = 111
32	12:00:40	SELECT * FROM bank_transactions LIMIT 0, 1000
33	12:00:40	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000
34	12:00:40	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000
35	12:00:40	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000

Info Session